

Bayesian Techniques in Macroeconomics

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Master in Quantitative Economics, second year

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CHAPTER 4

Bayesian Estimation of DSGE Models

Content of the Handout

1	A Bayesian perspective.....	p. 3
2	Bayesian estimation with Dynare	p. 9
	Exercise 1	p. 16
3	A full-fledged RBC model.....	p. 16
	Exercise 2	p. 22
	Exercise 3	p. 22

HANDOUT OBJECTIVES

- Estimating a DSGE model using Bayesian techniques;
- Decomposing business cycles into multiple sources of exogenous disturbances;
- Producing and evaluating forecasts.

Session roadmap

Economic intuition. Priors encode disciplined information about structural mechanisms; the likelihood confronts them with data. **Bayesian intuition.** The posterior combines both sources and MCMC

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quantifies uncertainty beyond the mode. **Model and data.** Check support, measurement equations, identification, and independent shocks. **MATLAB/Dynare exercise.** Estimate, inspect acceptance and mixing, and report posterior IRFs with credible intervals. **Takeaway.** Posterior claims require reproducible convergence and sensitivity diagnostics.

1. A Bayesian perspective

We estimate a state-space linear model using Bayesian techniques as in [Fernandez-Villaverde and Rubio-Ramirez \[2004\]](#) and [Smets and Wouters \[2007\]](#). Non-calibrated parameters are estimated using the Bayesian maximum likelihood principle (see [An and Schorfheide \[2007\]](#) for an extensive presentation). As exemplified by [Smets and Wouters \[2003\]](#) and [Smets and Wouters \[2007\]](#), this approach can fruitfully be applied to macroeconomic models as it allows a DSGE model to perform as well as an unrestricted VAR model in explaining macroeconomic time series. In our estimation we follow closely each of the steps defined by [Smets and Wouters \[2003\]](#).²

1.1 Building the likelihood function

Recall that in the first handout, any macroeconomic model with expectations can be cast in a compact form as follows:

$$\mathbb{E}_t [f_\Theta (x_{t+1}, x_t, x_{t-1}, \epsilon_t)] = 0 \quad (1)$$

where f_θ are the non-linear relations across variables x_t , stochastic innovations ϵ_t , for a given set of structural parameters Θ . By applying linearization methods

By applying linearization methods on the system of non-linear equation, it is possible to get of solution of the form

$$\hat{x}_t = F\hat{x}_{t-1} + G\epsilon_t, \quad (2)$$

where elements in matrices F and G are function of structural parameters Θ .

Let H be the selection matrix of observable variables – a subset of $\hat{x}_t$ – such that endogenous variables reads as $y_t = H\hat{x}_t + v_t$ with v_t the measurement errors. Then it is possible following the application of a filter (e.g. Kalman, inversion, particle,...) to get the errors of prediction:

$$S_t = Y_t - E_{t-1}\{y_t\},$$

where $E_{t-1}\{y_t\}$ is an output from the statistical filter and Y_t is a sample with n observed variables and T periods.

The errors of predictions of the estimated model follows $S_t \sim \mathcal{N}(\mu_t, \Omega_t)$ with μ_t the mean and Ω_t the variance-covariance of errors of predictions, the general expression of the likelihood function reads as follows:

$$\log \mathcal{L}(\theta | Y_T, \dots, Y_1) = -\frac{nT}{2} \log(2\pi) - \frac{1}{2} \sum_1^T \log(|\Omega_t|) - \frac{1}{2} \sum_1^T S_t' \Omega_t^{-1} S_t. \quad (3)$$

²The successive steps of this estimation strategy are presented by [Smets and Wouters \[2003\]](#) as follows: “this econometric approach attempts to provide a full characterisation of the observed data series. For example, following [Sargent \[1989\]](#), a number of authors have estimated the structural parameters of DSGE models using classical maximum likelihood methods. These maximum likelihood methods usually consist of four steps. In the first step, the linear rational expectations model is solved for the reduced form state equation in its predetermined variables. In the second step, the model is written in its state space form. This involves augmenting the state equation in the predetermined variables with an observation equation which links the predetermined state variables to observable variables. In this step, the researcher also needs to take a stand on the form of the measurement error that enters the observation equations. The third step consists of using the Kalman filter to form the likelihood function. In the final step, the parameters are estimated by maximising the likelihood function. Alternatively within this strong interpretation, a Bayesian approach can be followed by combining the likelihood function with prior distributions for the parameters of the model, to form the posterior density function. This posterior can then be optimised with respect to the model parameters either directly or through Monte-Carlo Markov-Chain (MCMC) sampling methods.”

The aim is to estimate the vector of parameters θ , a subset of all the structural parameters of a model $\theta \in \Theta$. Finally, in most software, it is easier to minimize than maximize, so the maximum-likelihood estimator reads as follows:

$$\hat{\theta}_{MLE} = \arg \min_{\{\theta\}} \{-\log \mathcal{L}(\theta|Y_T, \dots Y_1)\} \quad (4)$$

Note that errors of predictions S_t are actually a function of estimated parameters θ , i.e. some parameters values may improve the forecasting performance of the model, and thus reduce the volatility of the errors of predictions Ω_t . So the idea is to pick a value of $\hat{\theta}_{MLE}$ that maximize the joint densities.

1.2 Adding prior information

The Bayesian approach consists of defining an *a priori* distribution $p(\theta)$ over the structural parameters θ , including both a prior value and a distribution. The Bayesian estimate of a parameter thus mixes the prior mean with the maximum likelihood estimate. The choice of the initial values for the prior parameters requires an intuition for their role in the economy. We follow the standard practice of the literature in choosing one of the following probability distributions for parameters: the Normal distribution ($\mathcal{N}$), the Beta distribution ($\mathcal{B}$, for parameters with values lying between 0 and 1), the Gamma distribution ($\mathcal{G}$, for priors which allow relatively diffuse and positive support) and the Inverse Gamma distribution ($\mathcal{IG}$, for shocks, as this distribution allows us to choose priors that are strongly diffuse and positive).

The estimates of the parameters are updated using the dataset Y according to Bayes' rule:

$$p(\theta|Y) = \frac{p(\theta|Y)p(\theta)}{p(Y)} \approx L(\theta|Y)p(\theta), \quad (5)$$

where $L(\theta|Y)$ denote the likelihood function, $p(\theta|Y)$ is the posterior distribution of parameters and $p(Y)$ is the marginal likelihood (marginal data density).

Obtaining the analytical expression for the likelihood function is generally not possible. However, the posterior distribution of the model parameters can be constructed numerically by applying the Markov Chain Monte Carlo (MCMC) methods. Finally, we compute the posterior moments of the parameters using a sufficiently large number of draws, having made sure afterwards the MCMC algorithm has converged.

$$\hat{\theta}_{BE} = \arg \max_{\theta} L(\theta|Y)p(\theta) \quad (6)$$

One main advantage of DSGE models compared to VARs concerns the evaluation of monetary policy. DSGE models replicate time series by the responses of agents to different sequences of shocks. Once we have estimated all the sequences of shocks and the reaction parameters of households, firms and monetary authorities, we can perform counterfactual analyses by changing the parameters of monetary policy. Counterfactual scenarios can be derived from the estimated model by optimizing the policy coefficients in the Taylor to see how well the economy would have performed under an alternative policy regime.

1.3 An illustration with the RBC filter

Generating artificial series. Consider our 3 equation RBC model estimated with maximum likelihood and inversion filter (we could have done the same with the Kalman filter, but it does not fit into one single MATLAB file).

Recall that in the previous handout we had done the following:

- Simulating a model for 200 periods, we next assumed that only consumption was observable, and so we used consumption to retrieve the sequence of productivity shocks that generated the data.
- The distribution of productivity errors is given by $\varepsilon_t \sim \mathcal{N}(0, Q)$, let us assume that we want to estimate Q , i.e. the variance(-covariance) of errors. The true value of Q is $Q = 1$.
- The likelihood function is given by

$$\log L(\hat{Q}|\varepsilon_T, \dots, \varepsilon_1) = -\frac{nT}{2} \log(2\pi) - \frac{1}{2} \sum_1^T \log(|\hat{Q}|) - \frac{1}{2} \sum_1^T \varepsilon_t' \hat{Q}^{-1} \varepsilon_t.$$

- The Maximum Likelihood estimation is given by the value of Q that minimizes $-\log L$.

Setting prior information. Now we can introduce Bayesian econometrics. Which prior can we set for Q ? The best distribution would be one with positive support, but for clarity we choose the simplest distribution: the Gaussian distribution. Dynare, and most of macroeconomic papers, express any distribution into two moments: the mean and standard deviation of the distribution. Since a Gaussian distribution is already expressed in terms of its mean and standard deviation, it is straightforward to parameterize the prior.

Let us assume that the econometrician has some information about the distribution of Q and set as initial value $\hat{Q} \sim \mathcal{N}(1, 0.10)$. The joint probability density of observing our data (i.e. the log-likelihood function) is now augmented by the probability density of some estimated parameters, that we refer to as priors, denoted $P(\theta)$. Since our prior is $\hat{Q} \sim \mathcal{N}(1, 0.10)$, the log-likelihood function now reads as follows:

$$\underbrace{\log L(\hat{Q}|\varepsilon_T, \dots, \varepsilon_1)}_{\text{likelihood}} + \underbrace{\log p(\hat{Q}, 1, 0.10)}_{\text{priors}}$$

where $p(\cdot)$ is a Gaussian density with mean 1 and standard deviation 0.10. This translates into MATLAB as:

```
1 mu_Sig = 1;
2 sd_Sig = .1;
3 lnprior = @(x) log(normpdf(x,mu_Sig,sd_Sig));
```

Let us now explore the topology of the prior function in [Figure 1](#). We can see that $p(\hat{Q}, 1, 0.10)$ is higher when $\hat{Q}$ gets closer to the prior mean 1, so the prior information curbs the likelihood function in order to take some knowledge from the econometrician about the set of estimated parameters θ .

Optimizing the posterior density. Like for maximum likelihood, we must optimize an objective function. Now this objective function is the posterior density: the latter is proportional to the product of the prior density and the density of the sample (i.e. the likelihood function). This posterior distribution takes into account both the contribution of the data (i.e. the log-likelihood) and the prior information. We solve numerically the following minimization problem to obtain the mode of the posterior distribution:

$$\hat{Q}_{BE}^{\text{mode}} = \arg \min_Q -\log L(Q|Y) - \log p(Q) \quad (7)$$

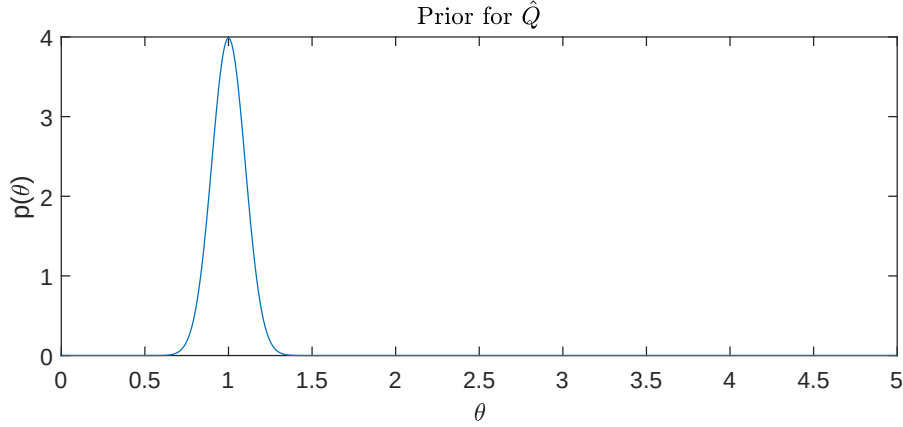


Figure 1: Prior distributions of variance covariance Q

```

1 theta0 = [0.01];
2 theta_MLE = fmincon(@(x) -llk(x),theta0);
3 theta_BE = fmincon(@(x) -(llk(x) + lnprior(x)),theta0);
4 disp('Mode:')
5 disp(['MLE: Est. sqrt(Q) ' num2str(sqrt(theta_MLE))])
6 disp(['BE : Est. sqrt(Q) ' num2str(sqrt(theta_BE))])

```

We call this point the distribution's mode because it is a local maximum of the posterior density. Having obtained the mode, we also need the mean of the density as well as the dispersion of the density. Problem: there is no closed-form expression to draw the density of the posterior distribution of Q . So we must use numerical techniques also needed to evaluate the posterior. Economists typically use Markov Chain Monte Carlo (MCMC) methods – in particular the Metropolis–Hastings algorithm – for this purpose.

Drawing uncertainty through Monte Carlo methods. The main purpose of the MH algorithm is to explore the parameter space of Q by weighting the outcomes appropriately. A Markov chain is a stochastic model describing a sequence of possible events in which the probability of each event depends only on the state attained in the previous event.

1. **Get covariance of estimated parameters.** We first need the variance-covariance-matrix of θ with respect to the posterior density. There are different ways to obtain it. In this example, we obtain it from the inverse of the Hessian matrix as in Dynare:

$$\Sigma_{\theta} = \left(\frac{\partial [\log L(\theta|Y) + \log p(\theta)]}{\partial \theta} \right)^{-1}, \quad (8)$$

2. **Markov Chain initialization.** We initialize each Markov Chain by setting the first starting candidate of the chain to the estimated posterior mode $\theta_0^{cand} = \hat{\theta}_{BE}^{mode}$. Next, we explore $i = \{1, \dots, N\}$ different random draws that lie in the neighborhood of $\hat{\theta}_{BE}^{mode}$.
3. **Draw a new candidate.** We next draw for each $i = \{1, \dots, N\}$ a new candidate θ_i^{cand} based on a random Gaussian perturbation of variance $c \times \Sigma_{\theta}$:

$$\theta_i^{cand} \sim \mathcal{N}(\theta_{i-1}^{cand}, c \times \Sigma_{\theta}) \quad (9)$$

Note that c is a scale parameter. It critically determines how large is the exploration space. If c is “too” small, the explored parameter space will be too narrow in the neighborhood of $\hat{\theta}_{BE}^{\text{mode}}$. Conversely, if the variance of perturbations is too big, the chain’s candidate θ_i^{cand} might get stuck into a lower local minimum. I discuss how to pick a not “too-big-too-small” c value, based a criterion called the acceptance rate that I discuss hereafter.

4. **Compute the acceptance probability of the candidate.** Is this new candidate θ_i^{cand} more favored by the data than the previous candidate $\theta_{i-1}^{\text{cand}}$? We can compute the acceptance rate as follows:

$$\alpha_i = \min \left(\frac{L(\theta_i^{\text{cand}}|Y) \times p(\theta_i^{\text{cand}})}{L(\theta_{i-1}^{\text{cand}}|Y) \times p(\theta_{i-1}^{\text{cand}})}, 1 \right). \quad (10)$$

Note that if α_i is high (i.e. close but below one), then the new candidate is favored while if α_i is low, then the initial candidate is favored.

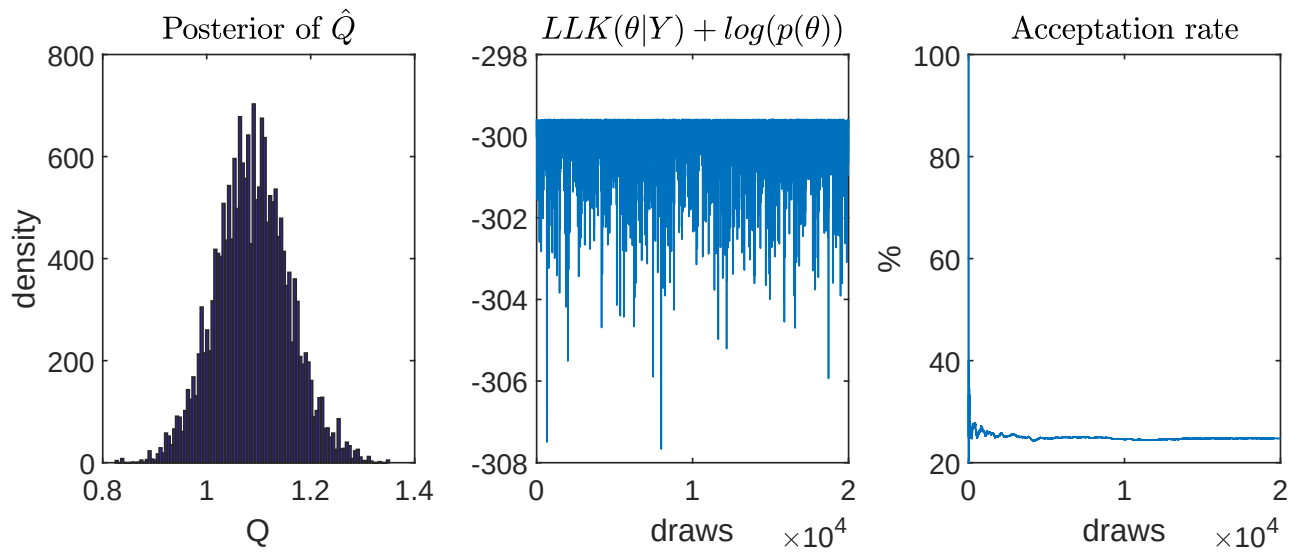
5. **Accept or reject the candidate.** Note that we are not trying to find the best value of θ (we already have it, it’s the mode of the posterior distribution), so the decision to accept or reject must also include a noise in order to accept candidates with low probability.

$$\theta_i^{\text{cand}} = \begin{cases} \theta_i^{\text{cand}} & \text{for } \mathcal{U}(0, 1) < \alpha_i \text{ (accept)} \\ \theta_{i-1}^{\text{cand}} & \text{(reject otherwise)} \end{cases}$$

Note that it is thus possible to accept a candidate with a low acceptance probability α_i , however, most likely candidates will emerge as the number of draws i increases.

6. **Repeat N times step 3, 4 and 5.**
7. **Truncate initial values.** Truncate a fraction of the first draws (to start to the n^{th} draw) to make sure the chain does not depend on the initial value θ_0^{cand} . The Markov chain now reads as $\{\theta_i^{\text{cand}}\}_{i=n}^N$.
8. **Analysis.** Analyse the posterior distribution using the histogram of $\{\theta_i^{\text{cand}}\}_{i=n}^N$. Compare the prior with its posterior distribution. Check also that the average acceptance rate, $E(\alpha_i)_{i=n}^N$, lies between 20% to 35%:
 - (a) if $E(\alpha_i)_{i=n}^N < 0.2$, the explored parameter space is too narrow and does not capture the true uncertainty stemming from the data. In that case, all the parameters will appear (erroneously) well identified.
 - (b) if $E(\alpha_i)_{i=n}^N > 0.35$, the explored parameter space is too large. The posterior distributions will include parameter sets that are rejected by the data. Some parameters may appear weakly identified while they are actually accurately identified.

MATLAB box 1 `bayesianRBC.m`



Shape	Name	Support	Example(s)
<code>normal_pdf</code>	$\mathcal{N}(\mu, \sigma)$	$(-\infty, +\infty)$	Policy parameters, utility curvatures,
<code>gamma_pdf</code>	$\mathcal{G}(\mu, \sigma)$	$(0, +\infty)$	Elasticities, utility curvatures, trends
<code>beta_pdf</code>	$\mathcal{B}(\mu, \sigma)$	$[0, 1]$	Probabilities, AR-MA terms, shares...
<code>inv_gamma_pdf</code>	$\mathcal{IG}_1(\mu, \sigma)$	$(0, +\infty)$	Standard deviations
<code>uniform_pdf</code>	$\mathcal{U}(\mu, \sigma)$	$(-\infty, +\infty)$	Same as normal distribution but with lower/upper bounds
<code>weibull_pdf</code>	$\mathcal{W}(\mu, \sigma)$	$[0, +\infty)$	For standard deviations, in particular of measurement shocks
<code>inv_gamma2_pdf</code>	$\mathcal{IG}_2(\mu, \sigma)$	$(0, +\infty)$	Standard deviations

Table 1: Prior distributions available in Dynare

2. Bayesian estimation with Dynare

Dynare also includes a quick and easy way to estimate linear DSGE models with the Kalman filter and prior information. Most of the code is going to be very similar to maximum likelihood estimation from the previous handout. The only difference lies in the specification of priors, and how from the maximization of the posterior density, we obtain the posterior distribution.

2.1 Priors

When we are estimating with Bayesian techniques, we need to specify prior distributions for each parameter to be estimated. In the estimated params section, for each estimated parameter, we declare a distribution, and specify parameters relevant for that distribution. The syntax for this is:

```
[parameter_name], [initial_value (optional)], [lower_bound (optional)], [upper_bound (optional)],
[prior_shape],
[prior_mean], [prior_standard_deviation], [p_3 (optional)], [p_4 (optional)], [jscale (optional)];
```

Table 1 details the distributions. Prior specification (in terms of shape, mean and standard deviation) allows to give more information about the distribution of the parameters and to avoid some regions where the likelihood function could be flat.

To illustrate the guideline for choosing a prior, let us consider first the AR(1) term of the productivity shock process ρ^A . To have a stationary model, this parameter must be below 1, while it must be above 0 to obtain a smooth response. The beta distribution meets this requirement as it has a support on $[0, 1]$ and will not allow this parameter to be above 1 or below 0. We therefore assume $\rho^A \sim \mathcal{B}(0.5, 0.2)$. In `estimated_params` block, we can add the following prior distribution as follows:

```
1 estimated_params;
2 rho_A, .8, 0, 1, beta_pdf, .5, .2;
3 end;
```

In the previous specification, we assume that each parameter is likely to have a mean of 0.5 and a standard deviation of 0.20. This prior is quite uninformative, as the support of this distribution allows the

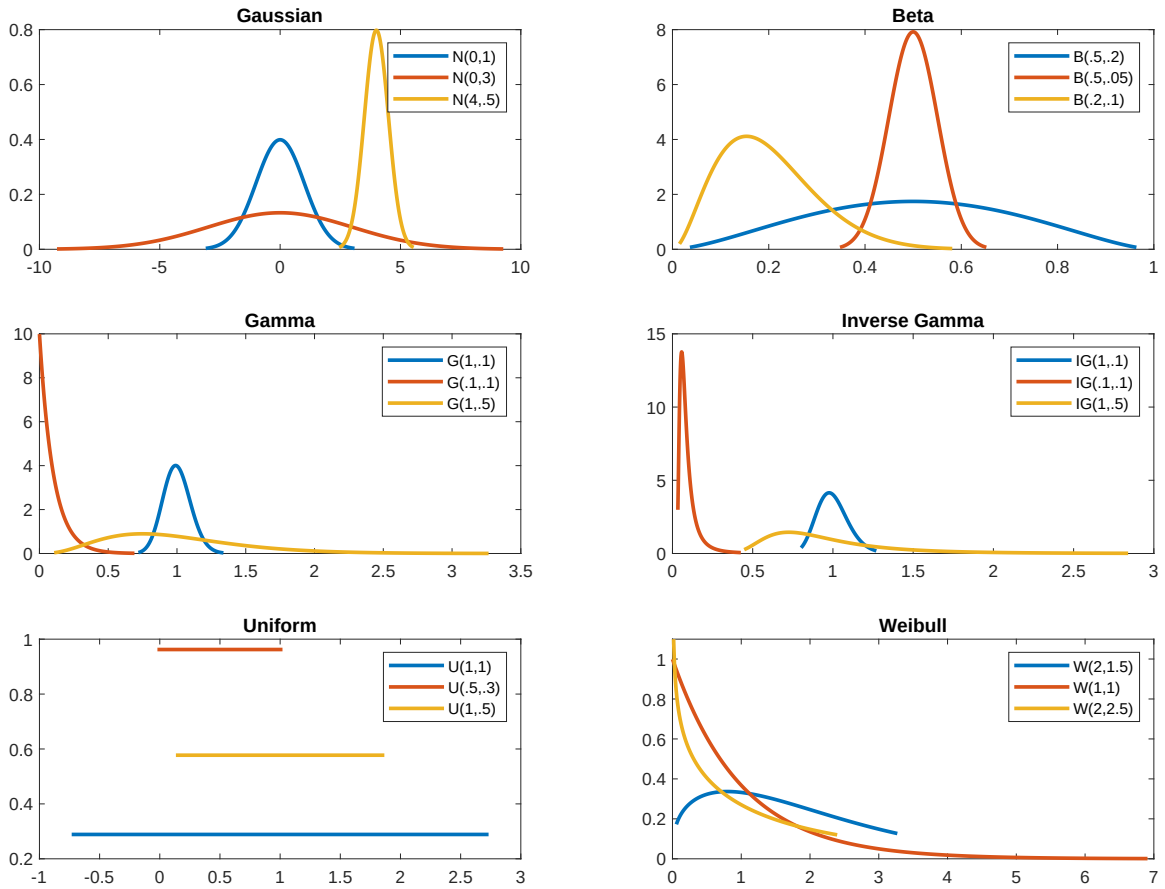


Figure 2: Prior distributions

parameter to be estimated near 1 or near 0. Conversely, an informative prior with a mean of 0.90 and a standard deviation of 0.01 would keep the estimate close to 0.90. Economists choose how informative their priors should be. For instance, if monetary policy shocks are known not to be strongly autocorrelated, a prior mean of 0.20 with a standard deviation of 0.05 could be fairly acceptable. Even through uninformative priors are preferred, it is possible to set tight priors but they have to be carefully justified.

Let us consider another example. The estimated standard deviation of a shock has to be positive, this constraint excludes *de facto* the Normal distribution as it has a negative support. A Beta distribution could be motivated as we want estimated shocks to be small otherwise our first order approximation becomes irrelevant with large shocks. However, demand and supply shocks may have a standard deviation higher than 1, which excludes the Beta distribution. It is then possible to use a Uniform distribution, as well as a Gamma or Inverse Gamma distributions. Here, we follow [Smets and Wouters \[2007\]](#) using an Inverse Gamma distribution with a mean of 0.10 and a standard deviation of 2 such that $\sigma^A \sim \mathcal{IG}(0.1, 2)$. Accordingly, we add in the Dynare file the lines which code the estimation of the shocks' volatility.

```
stderr
↪ e_a, .7, 0, inf, inv_gamma_pdf, .1, 2;
```

2.2 Estimation and posteriors

Once the priors are written in Dynare, we have to declare which variables are observable (and they should appear in the data file as well). Here, we estimate using one time series:

```
1 varobs      y;
```

Finally, we declare the estimation command as we did for maximum likelihood estimations. There are lots of options for Bayesian estimation which can go here; it's worth seeing the Dynare Manual for an overview of these options, and resources on Bayesian estimation for what they mean.

```
1 estimation(datafile=mydata,first_obs=1,mode_compute=4,
2 mh_replic=20000,mh_jscale=0.5,prefilter=1,lik_init=2,bayesian_irf,irf=30) y_obs pi_obs
  ↪ r_obs;
```

Here, `mh_replic` denotes the number of draws sampled by Metropolis-Hastings algorithm which aims at drawing the posterior distributions of estimated parameters. Increasing the number of draws helps the chains converge; nonetheless, it is not a sufficient condition for convergence. For small models, the number of draws can be low (roughly here 50 000 draws is fine); for large-scale models, it is preferable to have a larger number of draws to get converging chains (for instance, [Smets and Wouters \[2007\]](#) draws 200 000 draws). Option `lik_init` set to 2 allows Dynare to handle non-stationary observable variables in the fit exercise, we know that the GDP is a non-stationary variable requiring a special treatment for the estimation. Option `prefilter` remove the empirical mean in the observable matrix (i.e. it demeanes our sample). Estimating theoretical means would require adding a trend component in the production function, which we don't do here. Finally, option `mh_jscale` is a parameter that must be adjusted in order to obtain an acceptance ratio between 20% and 30% (if the final acceptance rate is below 20%, decrease `mh_jscale`; if it is above 30%, increase it). The interval for the acceptance ratio relies on research in Bayesian statistics that is beyond the scope of this handout.

Starting the estimation exercise, Dynare loads our data file, and starts to find the posterior mode using the `csmnwel` algorithm. The initial value of the log-likelihood function is `540.6002`, the algorithm employed by Dynare aims at maximising the value of the log-likelihood function by adjusting the estimated parameters.

console response

Initial value of the log posterior (or likelihood): 540.6002

f at the beginning of new iteration, -540.6001819383

Predicted improvement: 450.236373200

lambda = 1; f = -540.5959836

lambda = 0.33333; f = -761.5328209

lambda = 0.64439; f = -851.6090736

Norm of dx 0.30008

—

At the final iteration of the algorithm (here 58 iterations), the improvement performed in the last iteration is low enough for the algorithm to stop the optimisation. The value of the log-likelihood function is now **-895.6**. Dynare then prints in the console the mode of the posterior distribution:

console response

RESULTS FROM POSTERIOR ESTIMATION

parameters

prior mean mode s.d. prior pstdev
rho_A 0.500 0.9842 0.0085 beta 0.2000

standard deviation of shocks

prior mean mode s.d. prior pstdev
e_a 0.100 1.1003 0.0454 invg 2.0000

After finding the mode, the Metropolis-Hastings algorithm is employed. For each chain, it samples the number of draws we specified in the `estimation()` command.

Here, we have two chains of 20 000 draws. At the end of the sampling, we will get 40 000 samples. Dynare will automatically drop a certain percentages of draws (the percentage of drops can be manually set in the estimation command). Dynare compute the uncertainty (confidence intervals size can be manually set in the estimation command) and the mean of each parameter using the sequence of random samples from Metropolis-Hastings output.

console response

Log data density is 889.521825.

parameters

prior mean post. mean 90% HPD interval prior pstdev
rho_A 0.500 0.9817 0.9690 0.9946 beta 0.2000

standard deviation of shocks

prior mean post. mean 90% HPD interval prior pstdev
e_a 0.100 1.1064 1.0335 1.1844 invg 2.0000

Dynare prints the mean and 90% confidence interval upper and lower bounds as well as the prior standard deviation of the parameter. The posterior mean is preferred to the mode for empirical exercises. Dynare also runs some diagnostics to assess the convergence.

[Figure 3](#) reports some of the diagnostics computed by Dynare to examine the convergence of the MCMC chains as in [Brooks and Gelman \[1998\]](#). Here for instance, the red and blue lines should converge otherwise we should increase the number of draws, or adjust the priors/model.

[Figure 4](#) reports the prior and posterior distributions of parameters and shocks involved in the fit exercise. It is preferable to have a posterior distribution that differs from the prior. Having the posterior diverge from the prior means that your data (the likelihood in Bayes rule) is very informative about the parameter (if the posterior distribution is not too wide around the mode). If all your posterior estimates coincided with the prior mean, there would be no reason for estimation as the data does not add any new insights. In this situation, you should calibrate rather than estimate when the identification of your parameter is bad.

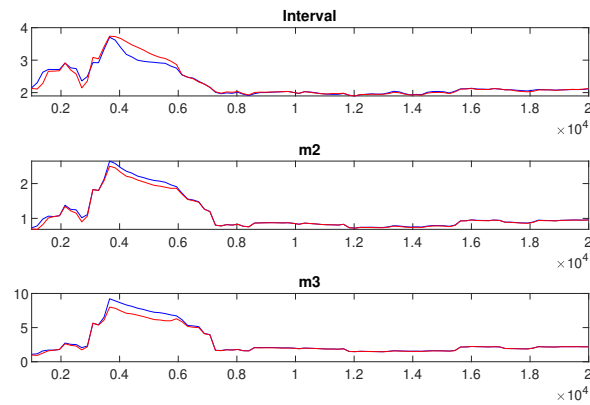


Figure 3: Univariate convergence diagnostic, Brooks and Gelman (1998)

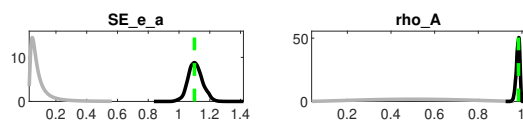


Figure 4: Priors (grey) and posteriors (black) distributions of estimated parameters and mean (green).

For instance in [Figure 4](#), ρ^A and σ^A are clearly well identified, [i.e.](#) data were informative for these parameters as their priors and posteriors are different. In our situation, we should decrease the mh acceptance rate.

Stored in	Description
oo_.posterior_mode	Parameters and shock standard errors at the posterior mode.
oo_.posterior_std	Standard error of the estimated values at the posterior mode.
oo_.posterior_density	Density of the posterior for each parameter (x,y values for each parameter).
oo_.prior_density	Density of the prior for each parameter (x,y values for each parameter).
oo_.SmoothedVariables	Estimated values of each endogenous variable for all time periods.
oo_.SmoothedShocks	Estimated values of each shock for all time periods.

Table 2: MATLAB's workspace after a Bayesian estimation

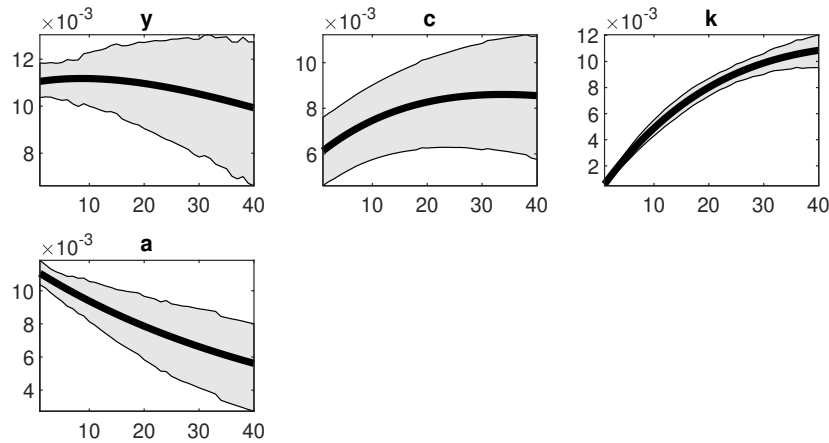


Figure 5: System response after an estimated supply shock

2.3 Bayesian system response

It is also worthwhile to have a look at the bayesian system response. [Figure 5](#) reports the system response of the model to an estimated supply shock. Following a productivity shock, physical capital is more productive, so households are more willing to postpone consumption and invest into physical capital to get more consumption in the future. More investment induces more demand for goods, which boosts GDP through both the demand for capital goods and by the increase in the stock of capital. The rise in income allows the households to increase immediately consumption. Note that the confidence interval is computed using Monte Carlo methods, but unlike VARs, it does not stem from the residual uncertainty, but originates from the parametric uncertainty.

2.4 Reloading results

Obtaining the posterior distribution can take quite a lot of time, that is why it is convenient if the model has not changed and if the posterior distribution is acceptable, then we can reload the results from the previous estimation as follows:

console response

```
estimation(datafile=mydata, first_obs=1, mh_replic=0, mh_jscale=0.5, prefilter=1, lik_init=2, load_mh_file, mode_file=NK3eq_estim_mode, mode_compute=0)
y y_obs pi_obs r_obs;
```

Exercise 1: Fixing the estimation

1. The acceptance rate is too high. Fix it.
2. Estimate the capital intensity parameter α . Use the same prior as [Smets and Wouters \[2007\]](#).
3. Why does the steady state must be updated?

3. A full-fledged RBC model

This model is inspired by the business cycle model of [Hansen \[1985\]](#) which aims at fixing anomalies in the real business cycle theory of [Lucas \[1977\]](#) and [Kydland and Prescott \[1982\]](#). Before Hansen, the intratemporal substitution between consumption and leisure was unable to explain large aggregate fluctuations in hours worked/employment.

3.1 The household

Our economy is populated by identical and infinite living households distributed over a continuum $j \in [0, 1]$. The representative household utility function reads as follows:

$$\mathcal{U}(C_{jt}, H_{jt}) = \frac{C_{jt}^{1-\sigma_C}}{1-\sigma_C} - \varepsilon_{t+\tau}^H \chi \frac{H_{jt}^{1+\sigma_L}}{1+\sigma_L}$$

Utility is increasing in consumption C_{jt} and decreasing in labor H_{jt} where σ^C and σ^L are curvature parameters while ε_t^H is a shock to the disutility of labor supply. Households in this economy maximise the expected value of their utility:

$$\max_{\{C_{jt}, H_{jt}, B_{jt}, K_{jt+1}, I_{jt}\}} \mathbb{E}_t \sum_{\tau=0}^{\infty} \beta^\tau \mathcal{U}(C_{jt}, H_{jt}), \quad (11)$$

where β is the discount factor. Households face an intertemporal problem, they find the value of consumption C_{jt} , hours worked H_{jt} and bonds B_{jt} which delivers the highest value of utility. They face the following budget constraint which binds every period:

$$w_t H_{jt} + z_t K_{jt-1} + r_{t-1} B_{jt-1} = C_{jt} + I_{jt} + S(I_{jt}, I_{jt-1}) + B_{jt} + T_{jt}, \quad (12)$$

where w_t is the real wage, z_t the remuneration of capital services, r_{t-1} is the real interest rate for bonds subscribed in $t-1$ and T_{jt} are lump-sum taxes used to finance the government budget. Here, $S(I_t, I_{t-1})$ is an investment cost function, which replicates congestion costs when firms are simultaneous investing at the same time. The functional form reads as:

$$S(I_{jt}, I_{jt-1}) = \frac{\kappa}{2} \left(\frac{I_{jt}}{I_{jt-1}} - 1 \right)^2 I_{jt-1} \quad (13)$$

where $\kappa \geq 0$ is an investment cost parameter.

The law of motion of capital also bounds the household's optimization problem:

$$K_{jt} = \varepsilon_t^I I_{jt} + (1 - \delta) K_{jt-1}, \quad (14)$$

where K_{jt} is the stock of capital bought in t (the lag is explained by the time-to-build hypothesis regarding capital setting, *i.e.* physical capital requires one period to be installed), I_{jt} denotes investment in t while parameter δ denotes the depreciation rate of physical capital. The lagrangian problem is given by:

$$\mathcal{L}_t = \mathbb{E}_t \sum_{\tau=0}^{\infty} \beta^\tau \left[+\lambda_{t+\tau}^c \left[\begin{aligned} & \frac{C_{jt+\tau}^{1-\sigma_C}}{1-\sigma_C} - \varepsilon_{t+\tau}^H \chi \frac{H_{jt+\tau}^{1+\sigma_L}}{1+\sigma_L} \\ & w_{t+\tau} H_{jt+\tau} + z_{t+\tau} K_{jt-1+\tau} + r_{t-1+\tau} B_{jt-1+\tau} \\ & -C_{jt+\tau} - I_{jt+\tau} - S(I_{jt+\tau}, I_{jt-1+\tau}) - B_{jt+\tau} - T_{jt+\tau} \end{aligned} \right] \right. \\ \left. + \lambda_{t+\tau}^k \left[\varepsilon_{t+\tau}^I I_{jt+\tau} + (1-\delta) K_{jt-1+\tau} - K_{jt+\tau} \right] \right]$$

where λ_t^c et λ_t^k are lagrange multipliers associated to the budget constraint and the physical capital accumulation constraint. The first order conditions are given by the welfare maximisation in C_{jt} , H_{jt} , I_{jt} , K_{jt} and B_{jt} :

$$\begin{aligned} (\partial C_{jt}) : \lambda_t^c &= C_{jt}^{-\sigma_C} \\ (\partial H_{jt}) : \varepsilon_t^H \chi H_{jt}^{\sigma_L} &= \lambda_t^c w_t \\ (\partial B_{jt}) : \mathbb{E}_t \left\{ \frac{\lambda_t^c}{\lambda_{t+1}^c} \right\} &= \beta r_t \\ (\partial I_{jt}) : -\lambda_t^c (1 + S'(I_{jt}, I_{jt-1})) + \beta \lambda_{t+1}^c S'(I_{jt+1}, I_{jt}) + \varepsilon_t^I \lambda_t^k &= 0 \\ (\partial K_{jt}) : \varepsilon_t^C \lambda_t^k &= \beta \mathbb{E}_t \left\{ \varepsilon_{t+1}^C \left[(1-\delta) \lambda_{t+1}^k + \lambda_{t+1}^c Z_{t+1} \right] \right\} \end{aligned}$$

Substituting the lagrange multipliers, our optimal conditions boils down to an Euler equation:

$$\beta r_t = \mathbb{E}_t \left\{ \left(\frac{C_{jt+1}}{C_{jt}} \right)^{\sigma_C} \right\}, \quad (15)$$

an hour supply equation:

$$W_t = \varepsilon_t^H \chi H_{jt}^{\sigma_L} C_{jt}^{\sigma_C} \quad (16)$$

It is possible to derive from the standard RBC model the endogenous asset price denoted $q_t = \lambda_t^k / \lambda_t^c$, defined as the ratio between the marginal cost of raising new capitals (Lagrangian multiplier on capital accumulation constraint λ_t^k) and the marginal utility of consumption (Lagrangian multiplier on budget constraint λ_t^c). Variable q_t also refers to the well known Tobin's Q published in [Tobin \[1969\]](#). Tobin's Q is the ratio between a physical asset's market value and its replacement value, this ratio has considerable macroeconomic implications as it constitutes the standard bridge between financial markets and markets for goods and services. In steady state, the ratio must be normalised to one to be consistent with asset pricing theory, such that $\bar{q} = 1$.

From its definition, we can thus substitute $\lambda_t^k = \lambda_t^c q_t$ in the two remaining first order conditions:

$$0 = -\lambda_t^c (1 + S'(I_{jt}, I_{jt-1})) + \beta \lambda_{t+1}^c S'(I_{jt+1}, I_{jt}) + \varepsilon_t^I \lambda_t^c q_t = 0 \quad (17)$$

$$\varepsilon_t^I q_t = (1 + S'(I_{jt}, I_{jt-1})) - \beta \frac{\lambda_{t+1}^c}{\lambda_t^c} S'(I_{jt+1}, I_{jt}) \quad (18)$$

$$\varepsilon_t^I q_t = 1 + \kappa \left(\frac{I_{jt}}{I_{jt-1}} - 1 \right) I_{jt-1} + \frac{1}{r_t} \frac{\kappa}{2} \left(\left(\frac{I_{jt+1}}{I_{jt}} \right)^2 - 1 \right) \quad (19)$$

It is straight forward to see that q_t is the real shadow value of capital goods is a non-linear function of the (expected) growth rate of investment. In absence of investment costs $\kappa = 0$ (and of the investment shock), the real value of physical capital goods becomes one.

While the first order condition on physical capital reads as:

$$\lambda_t^c q_t = \beta \mathbb{E}_t \left\{ \left[(1 - \delta) \lambda_{t+1}^c q_{t+1} + \lambda_{t+1}^c z_{t+1} \right] \right\} \quad (20)$$

$$q_t = \beta \mathbb{E}_t \left\{ \frac{\lambda_{t+1}^c}{\lambda_t^c} \left[(1 - \delta) q_{t+1} + z_{t+1} \right] \right\} \quad (21)$$

$$q_t = \frac{1}{r_t} \mathbb{E}_t \left\{ \left[(1 - \delta) q_{t+1} + z_{t+1} \right] \right\} \quad (22)$$

This equation is the no-arbitrage condition between risk free assets (i.e. the real interest rate) and risky assets (physical capital). The no-arbitrage condition is determined by the equality between the current cost of purchasing one additional unit of physical capital (left side) and the expected discounted return of this unit. $1/r_t$ reduces the expected return and features the opportunity cost of risk free assets.

3.2 Firms

Firms are homogeneous and distributed on a unit interval $i \in [0, 1]$ and have the following production technology:

$$Y_{it} = \varepsilon_t^A K_{it-1}^\alpha H_{it}^{1-\alpha} \quad (23)$$

where Y_{it} is the production, K_{it-1} and H_{it} are input factors, i.e. the quantity of physical capital and hours involved in the production process of firms. Finally, ε_t^A denotes the productivity shock process.

The optimisation problem can be expressed in two ways.

The first one can be expressed as a profit maximisation problem:

$$\max_{Y_{it}, K_{it-1}, H_{it}} Y_{it} - z_t K_{it-1} - w_t H_{it} + \lambda_t \left[\varepsilon_t^A K_{it-1}^\alpha H_{it}^{1-\alpha} - Y_{it} \right]$$

where λ_t denotes the lagrange multiplier on the supply constraint, it can also be interpreted as the marginal profit.

The second way can be viewed as an inputs minimisation problem:

$$\min_{K_{it}, H_{it}} Z_t K_{it-1} + W_t H_{it} - \lambda_t \left[\varepsilon_t^A K_{it-1}^\alpha H_{it}^{1-\alpha} - Y_{it} \right]$$

where λ_t denotes the lagrange multiplier on the supply constraint and can also be interpreted as the marginal cost of producing an additional unit of good.

First order conditions are:

$$\begin{aligned} (\partial K_{it-1}) : z_t &= \alpha \lambda_t \frac{Y_{it}}{K_{it-1}} \\ (\partial H_{it}) : w_t &= (1 - \alpha) \lambda_t \frac{Y_{it}}{H_{it}} \\ (\partial Y_{it}) : \lambda_t &= 1 \end{aligned}$$

Replacing the lagrange multiplier, the two conditions which determine the highest profits/lowest costs path are:

$$z_t = \alpha \frac{Y_{it}}{K_{it-1}} \quad (24)$$

$$w_t = (1 - \alpha) \frac{Y_{it}}{H_{it}} \quad (25)$$

3.3 Aggregation and Equilibrium conditions

Our economy is populated by two agents: households j and firms i . Their interactions, summarized by market clearing conditions for goods, labour and physical capital markets, can be expressed as follows:

$$\underbrace{\int_0^1 Y_{it} di}_{\text{aggregate supply}} = \underbrace{\int_0^1 C_{jt} dj}_{\text{aggregate consumption}} + \underbrace{\int_0^1 I_{jt} + S(I_{jt}, I_{jt-1}) dj}_{\text{aggregate investment}} + \underbrace{\bar{G} \varepsilon_t^G}_{\text{aggregate public spending}}$$

We assume output is driven exogenously by demand shocks, such that $\bar{G} = g^y \bar{Y}$, where g^y is the fixed spending-to-GDP ratio. Variable ε_t^G denotes the demand shock which affects output.

Letting Y_t , C_t and I_t denote the aggregate supply, consumption and investment, the resource constraint boils down to:

$$Y_t = C_t + I_t + S(I_t, I_{t-1}) + g^y \bar{Y} \varepsilon_t^G. \quad (26)$$

The equilibrium on the labour market is:

$$H_t = \underbrace{\int_0^1 H_{it} di}_{\text{labour demand}} = \underbrace{\int_0^1 H_{jt} dj}_{\text{labour supply}}, \quad (27)$$

and for physical capital:

$$K_t = \underbrace{\int_0^1 K_{it} di}_{\text{capital demand}} = \underbrace{\int_0^1 K_{jt} dj}_{\text{capital supply}}. \quad (28)$$

3.4 Disturbances

Our shock processes are characterized by four disturbances (in logs):

$$\log(\varepsilon_t^A) = \rho_A \log(\varepsilon_{t-1}^A) + \eta_t^A \quad \text{with } \eta_t^A \sim \mathcal{N}(0, \sigma_A^2), \quad (29)$$

$$\log(\varepsilon_t^G) = \rho_G \log(\varepsilon_{t-1}^G) + \eta_t^G \quad \text{with } \eta_t^G \sim \mathcal{N}(0, \sigma_G^2), \quad (30)$$

$$\log(\varepsilon_t^H) = \rho_H \log(\varepsilon_{t-1}^H) + \eta_t^H \quad \text{with } \eta_t^H \sim \mathcal{N}(0, \sigma_H^2), \quad (31)$$

$$\log(\varepsilon_t^I) = \rho_I \log(\varepsilon_{t-1}^I) + \eta_t^I \quad \text{with } \eta_t^I \sim \mathcal{N}(0, \sigma_I^2). \quad (32)$$

where in steady state these shocks are normalized to one.

3.5 Summary of the model

The model contains 13 equations and 13 endogenous variables:

$$\{r_t, C_t, H_t, Y_t, K_t, I_t, q_t, z_t, w_t, \varepsilon_t^A, \varepsilon_t^G, \varepsilon_t^H, \varepsilon_t^I\}.$$

$$\begin{aligned}
 \beta r_t &= \mathbb{E}_t \left\{ \left(\frac{C_{t+1}}{C_t} \right)^{\sigma_C} \right\} \\
 w_t &= \varepsilon_t^H \chi H_t^{\sigma_L} C_t^{\sigma_C} \\
 q_t &= \frac{1}{r_t} \mathbb{E}_t \left\{ [(1 - \delta)q_{t+1} + z_{t+1}] \right\} \\
 \varepsilon_t^I q_t &= 1 + \kappa \left(\frac{I_{jt}}{I_{jt-1}} - 1 \right) I_{jt-1} - \frac{1}{r_t} \frac{\kappa}{2} \left(\left(\frac{I_{jt+1}}{I_{jt}} \right)^2 - 1 \right) \\
 K_t &= \varepsilon_t^I I_t + (1 - \delta) K_{t-1} \\
 Y_t &= \varepsilon_t^A K_{t-1}^\alpha H_t^{1-\alpha} \\
 z_t &= \alpha \frac{Y_t}{K_{t-1}} \\
 w_t &= (1 - \alpha) \frac{Y_t}{H_t} \\
 Y_t &= C_t + I_t + S(I_t, I_{t-1}) + g^y \bar{Y} \varepsilon_t^G \\
 \log(\varepsilon_t^A) &= \rho_A \log(\varepsilon_{t-1}^A) + \eta_t^A \\
 \log(\varepsilon_t^G) &= \rho_G \log(\varepsilon_{t-1}^G) + \eta_t^G \\
 \log(\varepsilon_t^C) &= \rho_C \log(\varepsilon_{t-1}^C) + \eta_t^C \\
 \log(\varepsilon_t^I) &= \rho_I \log(\varepsilon_{t-1}^I) + \eta_t^I
 \end{aligned}$$

3.6 Linearization

The set of linearized equations reads as follows:

$$\begin{aligned}
 \frac{1}{\sigma_C} \hat{r}_t &= \mathbb{E}_t \{ \hat{c}_{t+1} \} - \hat{c}_t \\
 \hat{w}_t &= \hat{\varepsilon}_t^H + \sigma_L \hat{h}_t + \sigma_C \hat{c}_t \\
 \hat{q}_t &= \beta \mathbb{E}_t \left\{ [(1 - \delta)\hat{q}_{t+1} + \bar{Z}\hat{z}_{t+1}] \right\} - \hat{r}_t \\
 \hat{\varepsilon}_t^I + \hat{q}_t &= \kappa \bar{I} (\hat{i}_t - \hat{i}_{t-1}) - \beta \kappa (\hat{i}_{t+1} - \hat{i}_t) \\
 \hat{k}_t &= \delta (\hat{\varepsilon}_t^I + \hat{I}_t) + (1 - \delta) \hat{k}_{t-1} \\
 \hat{y}_t &= \hat{\varepsilon}_t^A + \alpha \hat{k}_{t-1} + (1 - \alpha) \hat{h}_t \\
 \hat{z}_t &= \hat{y}_t - \hat{k}_{t-1} \\
 \hat{w}_t &= \hat{y}_t - \hat{h}_t \\
 \hat{y}_t &= \frac{\bar{C}}{\bar{Y}} \hat{c}_t + \frac{\bar{I}}{\bar{Y}} \hat{i}_t + g^y \hat{\varepsilon}_t^G \\
 \hat{\varepsilon}_t^A &= \rho_A \hat{\varepsilon}_{t-1}^A + \eta_t^A \\
 \hat{\varepsilon}_t^G &= \rho_G \hat{\varepsilon}_{t-1}^G + \eta_t^G \\
 \hat{\varepsilon}_t^C &= \rho_C \hat{\varepsilon}_{t-1}^C + \eta_t^C \\
 \hat{\varepsilon}_t^I &= \rho_I \hat{\varepsilon}_{t-1}^I + \eta_t^I
 \end{aligned}$$

It is also common to estimate DSGE models on observable taken in growth rates. Therefore, letting ΔY_t denote the growth rate of output, the corresponding measurement equation reads as follows:

$$\begin{aligned}\Delta Y_t &= \log(Y_t/Y_{t-1}) \\ &= \log(Y_t) - \log(Y_{t-1}) + \log(\bar{Y}/\bar{Y}) \\ &= \log(Y_t/\bar{Y}) - \log(Y_{t-1}/\bar{Y}) \\ &\simeq \hat{y}_t - \hat{y}_{t-1}\end{aligned}$$

We estimate this model using output as an observable variable.

Exercise 2: Estimating the RBC model

Consider the code `fullRBC.mod` of the full-fledged RBC model.

1. Let us first include investment as an observable variable:
 - (a) Add a measurement equation in the model's code for the growth of investment di_t .
 - (b) Add a steady state for this variable.
 - (c) Add this observable to the `varobs`.
 - (d) We need to include another shock in the estimation. Which one should be pick up? Usually, we should include the shock that is more likely to generate fluctuations on this variable. Here, the investment shock is very likely to provide strong fluctuations on investment, so we are going to include ρ_i and $sd(\eta_{it})$ as parameters to estimate.
 - (e) Run the estimation.
 - (f) Investment adjustment cost κ may be estimated. Use the priors of [Smets and Wouters \[2007\]](#)
2. Examine the historical decomposition of shock for the observable variables.
3. Examine the forecasting error decomposition.
4. Let us now include consumption as observable. Same as for investment, include dc_t in the model's block, steady state, and set it in `varobs`. Use the government spending shock to estimate this series.
5. Finally, let us now include hours worked.
 - (a) This variable has no trend, so we don't need to apply the first difference filter. So we need to make our equations in Dynare consistent with the transformation of the data. Since the data are demeaned, the observable is the demeaned fraction of time per day devoted to work (the remaining part is the leisure). Include a new observable:

$$dh_t = h_t - \bar{h}$$
 - (b) Add the steady state of this variable.
 - (c) Add this variable to the observable block.
 - (d) Then estimate the remaining shock on the labor supply.
6. Some other structural parameters such as σ_C and σ_L may be estimated. Use the priors of [Smets and Wouters \[2007\]](#).

Exercise 3: External habits

Let us assume external consumption habits:

$$U(C_{jt} - hC_{t-1}, H_{jt+\tau}) \frac{(C_{jt} - hC_{t-1})^{1-\sigma_C}}{1-\sigma_C} - \varepsilon_{t+\tau}^H \chi \frac{H_{jt+\tau}^{1+\sigma_L}}{1+\sigma_L}$$

Note that C_{t-1} is the previous average consumption in the population, while C_{t-1} is agent specific. Habit Formation is based on catching up with the Joneses. The following first order conditions are affected:

$$\begin{aligned}(\partial C_{jt}) : \lambda_t^c &= U'_{c,t} \\(\partial H_{jt}) : \varepsilon_t^H \chi H_{jt}^{\sigma^L} &= \lambda_t^c w_t \\(\partial B_{jt}) : \mathbb{E}_t \left\{ \frac{\lambda_t^c}{\lambda_{t+1}^c} \right\} &= \beta r_t\end{aligned}$$

1. Update the first order conditions.
2. Find the steady state.
3. Linearize the equations.
4. Update the dynare files.

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